Trinomial Distribution

1 Definition

Suppose a random experiment is repeated n times independently and m is a prefixed positive integer; each repetition of the experiment may result in one of the three mutually exclusive events, say A_1, A_2 and A_3 with probabilities p_1, p_2 and $(1-p_1-p_2)$, respectively, where $0 < p_1, p_2 < 1$. Let X_i denote the number of events in A_i, $i = 1, 2, 3$, in n repetitions of the experiment. Then joint p.m.f of $(X_1, X_2)'$ is given by $-$

$$f_{\underline{X}}(x_1, x_2) = \frac{n!}{x_1! x_2! (n-x_1-x_2)!} p_1^{x_1} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2}, \begin{matrix} x_1 x_2 = 0, 1, \cdots, n; x_1 + x_2 \leqslant n; \\ 0 < p_1, p_2 < 1; p_1 + p_2 < 1. \end{matrix}$$

We say that the joint distribution of $(X_1, X_2)'$ is trinomial with parameters (n, p_1, p_2).

An example

Consider a box containing 20 balls, among which 8 are white, 7 are black, and 5 are red. Suppose 10 balls are drawn from the box one by one with replacement. If x_1 and x_2 denote respectively the number of white and red balls appearing in the sample, then their joint distribution is trinomial with parameter $\left(10, \frac{2}{5}, \frac{1}{4}\right)$.

2 Marginal Distribution

The marginal distribution of X_1 is $-$

$$\begin{aligned} f_1(x_1) &= \sum_{x_2=0}^{n-x_1} f_{\underline{X}}(x_1, x_2) \\ &= \binom{n}{x_1} p_1^{x_1} \sum_{x_2=0}^{n-x_1} \binom{n-x_1}{x_2} p_2^{x_2} (1-p_1-p_2)^{n-x_1-x_2} \\ &= \binom{n}{x_1} p_1^{x_1} (1-p_1)^{n-x_2}; x_1 = 0, 1, \ldots n; 0 < p_1 < 1. \end{aligned}$$

Hence, the marginal distribution of X_1 is binomial with parameters (n_1, p_1). Similarly, the marginal distribution of X_2 is in binomial with (n_2, p_2).

3 Conditional Distribution

The conditional distribution of X_2 given $X_1 = x_1$ is $-$

$$
\begin{aligned}
f_{2|1}(x_2 \mid x_1) &= \frac{f_{\underline{X}}(x_1, x_2)}{f_1(x_1)} \\
&= \frac{\dfrac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_1^{x_1} p_2^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2}}{\dfrac{n!}{x_1! (n - x_1)!} p_1^{x_1} (1 - p_1)^{n - x_1}} \\
&= \binom{n - x_1}{x_2} \left(\frac{p_2}{1 - p_1}\right)^{x_2} \left(1 - \frac{p_2}{1 - p_1}\right)^{n - x_1 - x_2}, x_2 = 0, 1, \ldots, n - x_1.
\end{aligned}
$$

Hence, the conditional distribution of X_2 given $X_1 = x_1$ is binomial with parameters $\left(n - x_1, \dfrac{p_2}{1 - p_1}\right)$. Similarly, the conditional distribution of X_1 given $X_2 = x_2$ is binomial with parameters $\left(n - x_2, \dfrac{p_1}{1 - p_2}\right)$.

This is intuitively obvious. Consider, those trials for which X_2 did not occur. There are ($n - x_2$) such trails, for each of which the probability that X_1 occurs is the conditional probability of X_1 given that X_2 has not occurred, i.e. $\dfrac{p_1}{1 - p_2}$.

4 Regression Equations and Correlation Coefficient

The regression equation of X_2 on X_1 is –

$$
E(X_2 \mid X_1 = x_1) = (n - x_1) \frac{p_2}{1 - p_1}
$$

The regression equation of x_1 on x_2 is –

$$
E(X_1 \mid X_2 = x_2) = (n - x_2) \frac{p_1}{1 - p_2}
$$

Hence, both the regression equations are linear with negative slopes. Now the regression coefficient of X_2 on X_1 is $\beta_{X_2 X_1} = -\dfrac{p_2}{1 - p_1}$ and the regression coefficient of X_1 on X_2 is $\beta_{X_1 X_2} = -\dfrac{p_1}{1 - p_2}$.

Hence, the correlation coefficient $\rho_{X_1 X_2}$ between X_1 and X_2 is –

$$
\rho^2_{X_1 X_2} = \beta_{X_2 X_1} \beta_{X_1 X_2} = \frac{p_1 p_2}{(1 - p_1)(1 - p_2)}
$$

Since the regression coefficients are negative, we get –

$$
\rho_{X_1 X_2} = -\sqrt{\frac{p_1 p_2}{(1 - p_1)(1 - p_2)}}
$$

Alternative,

$$
\begin{aligned}
E(X_1 X_2) &= E(E(X_1 X_2 | X_2 = x_2)) \\
&= E(X_2 E(X_1 | X_2 = x_2)) \\
&= E(X_2 (n - X_2) \frac{p_1}{1 - p_2}) \\
&= \frac{p_1}{1 - p_2} E(n X_2 - X_2^2) \\
&= \frac{p_1}{1 - p_2} \left(n^2 p_2 - n p_2 (1 - p_2) - n^2 p_2^2\right) \\
&= \frac{p_1}{1 - p_2} (n(n - 1) p_2 (1 - p_2)) \\
&= n(n - 1) p_1 p_2
\end{aligned}
$$

So,

$$\begin{aligned}\operatorname{cov}(X_1, X_2) &= E(X_1X_2) - E(X_1)E(X_2)\\ &= n(n-1)p_1p_2 - n^2p_1p_2\\ &= -np_1p_2\end{aligned}$$

Hence,

$$\begin{aligned}\rho(X_1, X_2) &= \frac{\operatorname{cov}(X_1, X_2)}{\sqrt{\operatorname{var}(X_1)}\sqrt{\operatorname{var}(X_2)}}\\ &= \frac{-np_1p_2}{\sqrt{n^2p_1p_2(1-p_1)(1-p_2)}}\\ &= -\sqrt{\frac{p_1p_2}{(1-p_1)\,(1-p_2)}}\end{aligned}$$

5 Joint Moment Generating Function

The joint MGF of $(X_1, X_2)'$ about the origin $(0, 0)$ is given by $-$

$$\begin{aligned}M_{\underline{X}}(t_1, t_2) &= E\left(e^{t_1X_1+t_2X_2}\right)\\ &= \sum_{x_1=0}^{n}\sum_{x_2-0}^{n-x_1} e^{t_1x_1+t_2x_2}\frac{n!}{x_1!x_2!\,(n-x_1-x_2)!}p_1^{x_1}p_2^{x_2}\,(1-p_1-p_2)^{(n-x_1-x_2)}\\ &= \sum_{x_1=0}^{n} e^{t_1x_1}\binom{n}{x_1}p_1^{x_1}\sum_{x_2=0}^{n-x_1}\binom{n-x_1}{x_2}\left(e^{t_2}p_2\right)^{x_2}(1-p_1-p_2)^{n-x_1-x_2}\\ &= \sum_{x_1=0}^{n}\binom{n}{x_1}\left(e^{t_1}p_1\right)^{x_1}\left(1-p_1-p_2+e^{t_2}p_2\right)^{n-x_1},\ \text{by using binomial theorem}\\ &= \left(1-p_1-p_2+e^{t_1}p_1+e^{t_2}p_2\right)^n\end{aligned}$$

Remark

Setting $t_1 = t_2 = t$ in $M_{\underline{X}}(t_1, t_2)$, we get the MGF of $X_1 + X_2$ $-$

$$M_{X_1+X_2}(t) = \left(1-p_1-p_2+(p_1+p_2)e^t\right)^n$$

which is the MGF of a binomial distribution with parameters $(n, p_1 + p_2\ )$. Hence, by the uniqueness property of MGF,

$$X_1 + X_2 \sim \operatorname{Bin}(n, p_1 + p_2)$$

6 Definition of Multinomial Distribution

Now, suppose that there are k outcomes possible at each of the n independent trails. Denote the outcomes $A_1, A_2, \ldots, A_K$ and the corresponding probabilities $p_1 \ldots, p_K$ where $\sum_{j=1}^{k} p_j = 1$. Let, X_j count the number of times A_j occurs,

Then,

$$\begin{aligned}&p\left(X_1 = x_1, \ldots X_{k-1} = x_{k-1}\right)\\ &= \frac{n_!}{x_1!x_2!\cdots x_{k-1}!\left(n-\sum_{j=1}^{k-1}x_j\right)!}p_1^{x_1}p_2^{x_2}\cdots p_{k-1}^{x_{k-1}}p_k^{n-\sum_{j=1}^{k-1}x_j}\end{aligned}$$

whore $x_1, x_2 \ldots, x_{k-1}$ are non-negative integers with $\sum_{j=1}^{k-1} x_j \leqslant n$

7 Exercises

1. Three coins are tossed 10 times. Let X_1 and X_2 denote, respectively, the number of times that no head appears and the number of times that exactly one head appears. Find −
   a) the joint distribution of (X_1, X_2).
   b) the conditional distribution of X_1 given $X_2 = 3$.
   c) the regression equation of X_1 on X_2.
   d) the correlation coefficient between X_1 and X_2.
   e) the conditional distribution of X_1 given $X_1 + X_2$.
   f) the distribution of $X_1 + X_2$.

2. If the joint distribution of $(X_1, X_2)'$ is trinomial with parameters (n_1, p_1, p_2). Find the point at which $f_{\underline{X}}(x_1, x_2)$ attains its maximum value.